Team 292

M2: Mid-submission — Green Finance Assistant

Megathon X: Singularity · PS1 — Green Customer Insights Platform (Lloyds Banking Group)

1. Current approach and changes since M1

The M1 architecture is unchanged and is now fully implemented as a working platform: strict separation of concerns — **ML predicts, a deterministic carbon engine calculates, the LLM only explains, and rule-based checks decide eligibility.** Delivered since M1: the synthetic dataset (seed 42: 120 fictional Indian customers, 13,110 transactions, 12-month windows, 8 mutually exclusive categories), the interpretable propensity engine with reason codes, the deterministic carbon engine with a documented factor registry, the grounded LLM explanation layer (streaming, with deterministic fallback), the Customer Green Companion and the Lloyds Intelligence Console.

Changes since M1, all driven by experiments:

- **Label rule refined:** synthetic take-up labels are now conditioned on eligibility — an ineligible customer cannot take up a product, so those rows are labelled 0. This removed impossible-take-up noise from the benchmark.
- **Model weights rebalanced once** after the first evaluation run showed the single-signal baseline beating the model on 3 of 6 products (details in Section 2).
- **Added beyond plan:** a customer-memory extension (re-orders recommendations from feedback without ever mutating propensity, eligibility or impact), a grounded companion chat answering only from structured data, an interactive judge-verification page exposing live recomputation, and a **CSV upload pipeline** that runs the entire engine — keyword-rule categorisation, documented profile-derivation heuristics, propensity, eligibility, carbon impact — on judge-provided transaction data in a sandbox, leaving the curated dataset untouched.

2. Experiments run so far and their results

Setup. 720 customer × product pairs scored; customer-level disjoint splits via sha256(customer_id) (train 432 / validation 120 / test 168 rows — no leakage); synthetic take-up labels from a documented, independent noisy rule (sigmoid of weighted signals + N(0, 0.5), seeded per pair) conditioned on eligibility; baselines: random (0.5) and the strongest single-signal heuristic per product. Metrics on the held-out test split; 95% bootstrap confidence intervals (1,000 seeded resamples).

Product	n (pos)	Model AUC	95% CI	Single-signal AUC	Random	P@k	Brier
EV Finance	28 (9)	**1.000**	[1.00, 1.00]	0.854	0.500	1.00	0.104
Paperless Banking	28 (9)	**0.918**	[0.79, 1.00]	0.927	0.500	0.75	0.186
Green Home Loan	28 (6)	**0.871**	[0.70, 1.00]	0.311	0.500	0.50	0.174
Green Personal Loan	28 (9)	**0.860**	[0.69, 0.98]	0.702	0.500	0.75	0.255
Green Investments	28 (7)	**0.830**	[0.64, 0.96]	0.561	0.500	0.50	0.200
Solar Rooftop Finance	28 (7)	**0.748**	[0.56, 0.90]	0.524	0.500	0.25	0.185
Macro average	168	**0.871**	[0.82, 0.93]	0.646	0.500	**0.625**	**0.184**

Reading the results. The model beats the single-signal baseline on 5 of 6 products and on macro average (0.871 vs 0.646). Paperless Banking is a deliberate honest near-tie (0.918 vs 0.927): digital-activity alone is nearly sufficient there, and we report rather than hide it. Solar is the weakest product (0.748) — its take-up signal is thinner at 120 customers; the wide CI says so explicitly.

Experiment — label conditioning. Before conditioning labels on eligibility, macro AUC was 0.694 and the model lost to the single-signal baseline on 3 products; inspection showed take-up labels being generated for ineligible customers. Conditioning on eligibility (a modelling decision, documented) raised macro AUC to 0.871. **Experiment — weight rebalance.** EV, Paperless and Green Investments weights were rebalanced once towards the multi-signal structure; per-product AUCs are reported after this single tuning pass, with no further p-hacking.

3. Progress on testing

- **Carbon engine:** 4/4 known-answer unit tests pass on every evaluation pass; 11/11 public test vectors reproduce documented outputs exactly; 6/6 featured-customer chains recompute end-to-end from raw record inputs with zero mismatch.
- **Guardrails (live DB):** ineligible products never recommended (0 violations); every recommendation carries evidence, confidence, reason codes, timestamp and an impact estimate (0 violations).
- **LLM layer:** fallback determinism (identical context → identical narration) and number-grounding (every figure in fallback narration appears in the structured context) both pass; live explanations observed citing exact engine figures.
- **Reproducibility:** seed-42 dataset regeneration matches customers, transactions and amount checksum; two consecutive full harness runs are byte-identical. Harness runtime ~250 ms.

- **Data quality:** 242 injected duplicates flagged and excluded from features; 100% categorisation coverage; complete 12-month windows.
- **External-data generality:** the pipeline was re-run on an independent CSV (3 customers, 384 rows, schema-validated with 0 rejections): it derived profiles from documented heuristics (income ≈ 2× monthly spend; vehicle inferred from fuel spend; home ownership from electricity draw), categorised 100% of rows via keyword rules, and detected 2 guardrail-compliant recommendations with reason codes and impact estimates (EV Finance propensity 0.62, Green Home Loan 0.38) — the same code path as the curated dataset.
- **Exposure:** all of the above is runnable by judges on a dedicated verification page (live engine tester, dummy input→output vectors, customer-level recompute, reproducibility proof).

4. Plan for the remaining time

- **Final report PDF** (before 11:00 AM): full results with the evaluation methodology above, plus the Responsible AI note — privacy, bias, explainability, hallucination control, human oversight and data limitations.
- **Demo video:** scripted walkthrough mapping each scoring criterion to a live screen, ending on the judge-verification MATCH verdict.
- **Code zip + slides** (slides by 11:30 AM): freeze features, final end-to-end run-through, package and push; all PDFs committed to deliverables/ on main ahead of deadlines.
- **No new features** are planned; remaining effort is documentation, rehearsal and freeze discipline.